

Resonant MEMS Mirror and Phase Locked Loop

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1 Introduction

Scanning mirrors based on micro-electro-mechanical systems (MEMS) technologies are one of the most promising solutions for various automotive applications, e.g. lidars and augmented reality head-up display, thanks to a small form factor, high scanning frequencies with a wide field of view, high contrasts and brightness, and competitive unit cost by mass production. To ensure stable operation MEMS scanning systems have to be resistant to environmental conditions such as vibrations, shocks, and temperature variations. In this laboratory exercise, the nonlinear dynamics, electrostatic comb drive actuation, and current self-sensing of a resonant MEMS mirror are investigated. Furthermore, a phase locked loop (PLL) is developed to enable feedback control of the MEMS mirror.

2 A Resonant MEMS Mirror and Phase Locked Loop

2.1 Concept of the MEMS Lidar with a 1D MEMS Mirror

Figure 1 illustrates the operational principle of a 1D scanning MEMS lidar. This technology employs a MEMS mirror for scanning along the horizontal axis while multiple lasers illuminate one full vertical line. A detector array captures the reflected light. This method concentrates laser power on specific image sections, enabling extended range due to increased intensity. By scanning lines rather than individual points, the 1D scanning MEMS lidar achieves a relatively high frame rate. Distance measurements rely on the measurement of the time of flight of short laser pulses generated by the lasers. With knowledge of the mirror angle, lasers can be activated at desired angles, enabling an array of distance measurements along the vertical line. Repeated across multiple scan lines, this process yields a 3D environmental scan. Precise driving of the MEMS mirror and sensing of its angle is imperative for obtaining accurate 3D scans.

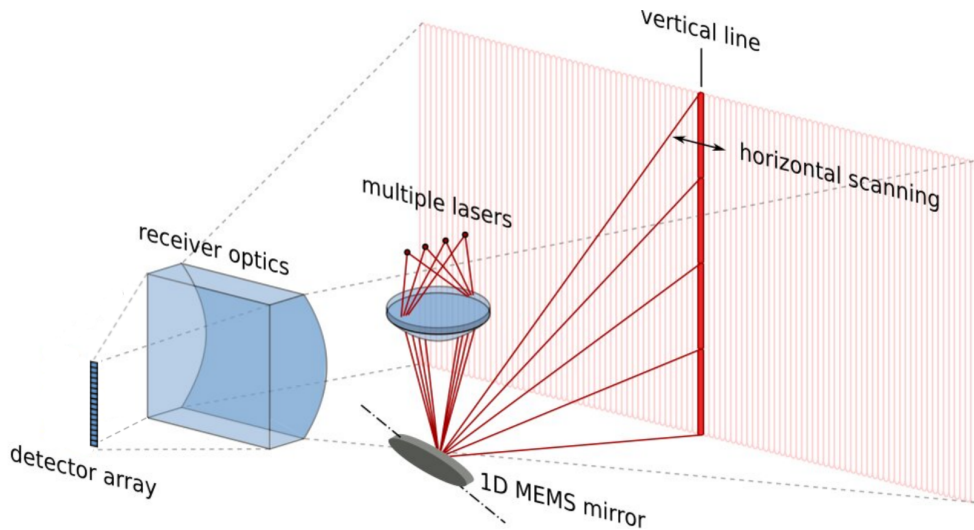


Figure 1: Conceptual diagram of 1D scanning MEMS lidar. [1].

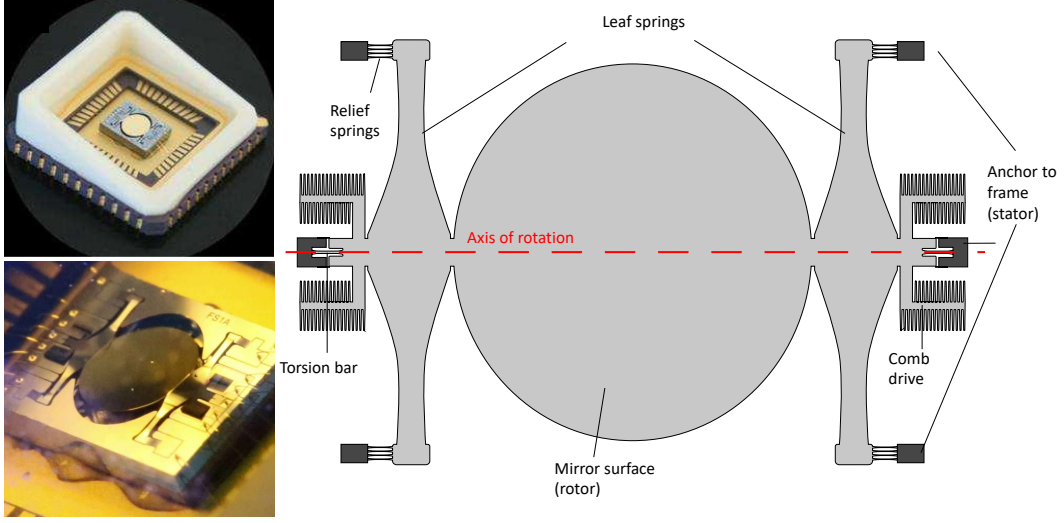


Figure 2: Pictures of the MEMS mirror and conceptual drawing.

2.2 A Resonant MEMS Mirror

Figure 2 shows pictures and a conceptual drawing of the MEMS mirror. The MEMS mirror consists of a rotating mirror surface held by torsion bars and leaf springs. Short and narrow torsion bars efficiently suppress rigid body modes different from the intended rotational motion. The leaf springs provide the dominant portion of the restoring torque, enabling high scanning frequencies. The mirror is actuated by comb drives, which due to the electrostatic forces between the charged plates of the capacitor generate the required torque.

2.2.1 Nonlinear Dynamics of the Resonant MEMS Mirror

The equation of motion for the MEMS mirror is given by

$$J\ddot{\theta} + c\dot{\theta} + (k_1 + k_3\theta^2)\theta = \tau_{\text{comb}}, \quad (1)$$

with the mirror angle θ , moment of inertia J , and damping c . The external torque τ_{comb} is produced by the comb drives. The stiffness is defined by a linear term k_1 and a positive nonlinear term k_3 . This type of nonlinear differential equation is called “Duffing equation”. The damping c of the mirror is small compared to the stiffness, which results in a high Q factor. Since $k_3 > 0$, the stiffness of the system, i.e. $(k_1 + k_3\theta^2)$, increases as the angle θ becomes larger, leading to an increase in the natural frequency f . This behavior is reflected as a bending of the backbone curve in Figure 3. The backbone curve indicates that the resonance frequency of the system changes if the amplitude changes, and represents the pure mechanical property without any actuation torque, i.e. $\tau_{\text{comb}} = 0$. The bottom and top response curve indicate the response with a square wave driving signal. Near its resonance, the system can be operated at the bottom response curve. From there, decreasing the frequency leads to an increase in mirror amplitude. At a specific frequency, an amplitude jump occurs and the amplitude increases drastically. The system is now operating at the top response curve, which also shows a different phase delay than the bottom response curve. From there, the frequency can be increased again to reach an even higher mirror amplitude. When the mirror amplitude is too high, the system falls back to the bottom response curve.

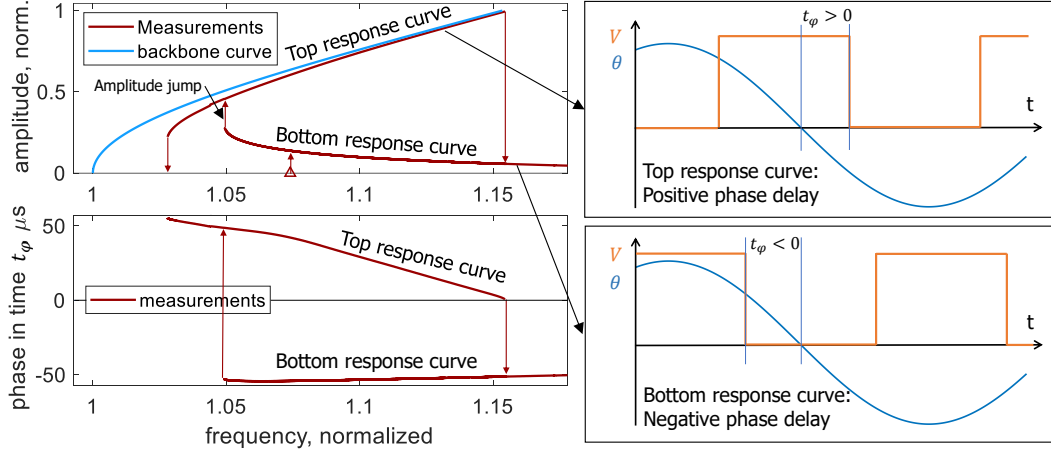


Figure 3: Normalized response curves of the mirror. The backbone curve shows the frequency response without actuation. Due to the nonlinear stiffness, the resonance frequency depends on the amplitude, resulting in a bending of the frequency response. The amplitude and phase between top response curve and bottom response curve, measured while a square wave driving signal is applied, differ.

To derive the backbone curve, let the damping c be 0, and the nonlinearity be weak, i.e. $k_3 \ll k_1$. Under these assumptions, the solution of (1) can be approximated by a single tone cosine as $\theta = a \cos(\omega t)$, with amplitude a and angular velocity $\omega = 2\pi f$. Applying this solution and $\tau_{\text{comb}} = 0$ to (1), and using the trigonometric identity $\cos^3(\omega t) = \frac{1}{4} (3 \cos(\omega t) + \cos(3\omega t))$ leads to

$$a \left(-J\omega^2 + k_1 + \frac{3}{4}k_3 a^2 \right) \cos(\omega t) + a^3 \left(\frac{3}{4}k_3 \right) \cos(3\omega t) = 0. \quad (2)$$

Since the functions of $\cos(\omega t)$ and $\cos(3\omega t)$ are orthogonal to each other, both coefficients have to be zero. Neglecting the terms of $\cos(3\omega t)$, because $k_3 \ll 1$, the backbone curve is approximated by setting the coefficient of $\cos \omega t$ to zero, resulting in

$$f(a) = \frac{1}{2\pi} \sqrt{\frac{k_1}{J} + \frac{3}{4} \frac{k_3}{J} a^2}. \quad (3)$$

For amplitude a close to zero f degenerates to the classical natural frequency of a linear mechanical system, i.e. $f(0) = \frac{1}{2\pi} \sqrt{\frac{k_1}{J}}$. The backbone curve can be measured by observing the frequency during the pure mechanical ring-down oscillation without actuation input.

The nonlinear stiffness $k_3 > 0$ mainly originates from the thin and wide leaf spring, which forms a doubly clamped beam, bending up and down by the mirror oscillation. The nonlinear stiffness enables a larger range of frequencies with high amplitude, allowing uniform operation frequency even if the natural frequency varies due to manufacturing tolerances.

2.2.2 Electrostatic Actuation by Comb Drives and Parametric Excitation

The comb drives consist of comb fingers on both the stator and the rotor side of the MEMS mirror, as shown in Figure 4. The comb drive fingers, i.e. a number of thin plates, form

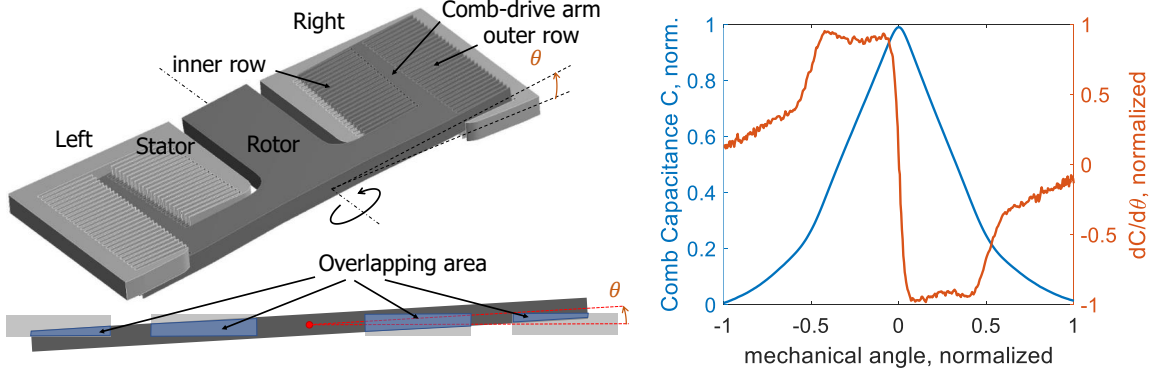


Figure 4: (left) Structure of the comb drive and overlap region according to the mirror angle. (right) capacitance and angular derivative of capacitance, measured from the resonant MEMS mirror.

a capacitor between the stator electrode. The area of the comb drive changes if the mirror rotates, therefore the capacitance $C(\theta)$ of the comb drive is angle dependent. The maximum area, and therefore also maximum capacitance, is reached when the comb drive fingers fully overlap at $\theta = 0$. When the comb drives fully disengage, the torque approaches zero. There is an inner row and an outer row of comb drive fingers. The outer row provides more torque, but only at small angles. Figure 4 shows the capacitance C and the derivative of the capacitance with respect to θ .

The torque produced by the electric field is the angular derivative of the energy stored in the capacitor as

$$\tau_{\text{comb}}(t) = \frac{d}{d\theta} \left(\frac{1}{2} C V_{\text{act}}^2(t) \right) = \frac{1}{2} \frac{dC}{d\theta} V_{\text{act}}^2(t), \quad (4)$$

where $V_{\text{act}}(t)$ denotes the time-dependent actuation voltage applied between the rotor and the stator of the comb drives. Since (4) contains V_{act}^2 , the direction of the torque does not depend on the sign of the driving voltage, but only on the sign of $\frac{dC}{d\theta}$. Figure 4 shows, that the torque is always positive if the θ is negative, and vice-versa, indicating that the comb drive can only pull the mirror towards the center position, acting like an additional spring. For implementation simplicity, the actuation voltage $V_{\text{act}}(t)$ is a square wave from 0 to peak voltage V_p . Since the comb drive acts like an additional spring if the driving voltage is on, the driving signal modules the total stiffness, i.e. a parameter of the system. Therefore, the electrostatic actuation by the comb drive can be characterized as “parametric excitation”.

2.2.3 Self-Sensing Concept by Displacement Currents

Figure 4 shows that the capacitance of the comb drive depends on the angle of the mirror. In case the square wave driving voltage is on, i.e. $V_{\text{act}}(t) = V_p$, the change in capacitance causes a change in charge, and therefore a flow of the displacement current. The angle of the mirror can be measured from the displacement current of the comb drive, contemporary to using the comb drive for actuation. Therefore, this principle is called “self-sensing”. Self-sensing allows simple and compact system design without the need for an additional sensor.

If the driving voltage is on, the displacement current is given by the change in charge of

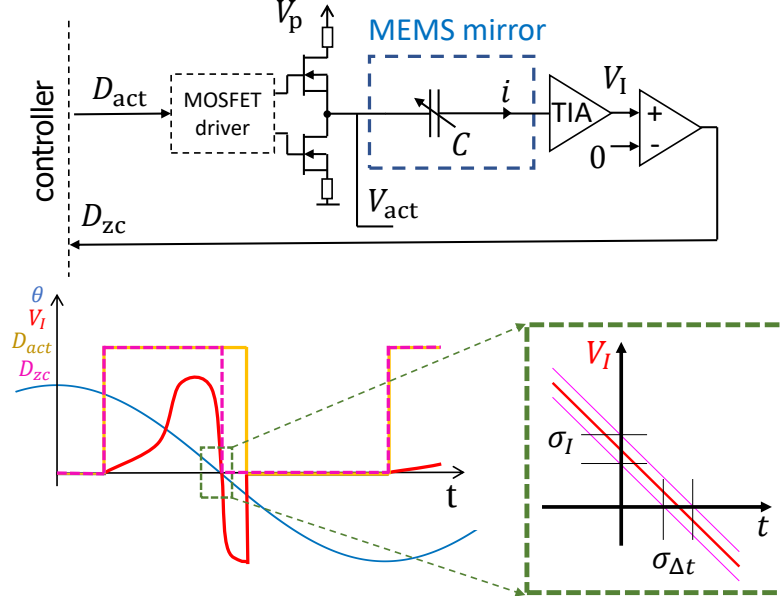


Figure 5: (Top) Concept diagram of the current self-sensing and (bottom) conceptual diagram of the signaling and an interpretation of the timing jitter $\sigma_{\Delta t}$ from the measurement noise σ_I [2].

the capacitance, i.e.

$$i = \frac{dQ}{dt} = \frac{d}{dt} (CV_p) = V_p \frac{dC}{dt} = V_p \frac{dC}{d\theta} \frac{d\theta}{dt}, \quad (5)$$

where $d\theta/dt$ is an angular velocity of the mirror, i.e. $\dot{\theta}$. Since $\frac{dC}{d\theta}$ shows a sharp transition when the mirror crosses 0° , and $\dot{\theta}$ is constant when $\theta \approx 0$, the current i also transitions quickly from positive to negative at 0° . Moreover, the current is 0, if the mirror angle is 0. This can be detected by a comparator, which provides a digital zero-crossing signal D_{zc} to the controller. The controller measures the time difference between the falling edge of D_{zc} and the falling edge of V_{act} to obtain the phase of the mirror in time. Note that this measurement only works when the driving voltage is turned on, i.e. the phase measurement by self-sensing is only possible for the operation on the top response shown in Figure 3.

The timing measurement, however, is accompanied by a timing jitter, caused by the noise of the current signal i . Assume that the displacement current measurement contains a Gaussian stochastic noise with a standard deviation of σ_i . Then the standard deviation of the timing jitter can be given by the slope of the current as shown in Figure 5.

$$\sigma_{\Delta t} = \frac{1}{|m|} \sigma_I, \quad (6)$$

where m is the slope of the displacement current at the zero-crossing of V_I , given by

$$m = \left. \frac{di}{dt} \right|_{\theta=0} \quad (7)$$

$$= V_p \left. \frac{d}{dt} \left(\frac{dC}{d\theta} \frac{d\theta}{dt} \right) \right|_{\theta=0} \quad (8)$$

$$= V_p \left. \frac{d^2 C}{d\theta^2} \left(\frac{d\theta}{dt} \right)^2 \right|_{\theta=0} + V_p \left. \frac{dC}{d\theta} \frac{d^2 \theta}{dt^2} \right|_{\theta=0} \quad (9)$$

$$= V_p \left. \frac{d^2 C}{d\theta^2} \right|_{\theta=0} \left(\max(\dot{\theta}) \right)^2 \quad (10)$$

The timing jitter represents the uncertainty of the timing of the mirror zero angle, leading to the uncertainty in the phase measurement. Since the laser triggering is timed by the phase measurement, the timing jitter harms the positioning of the laser trigger. The standard deviation of the optical angular errors can be given as $\sigma_{opt} = 2 \max(\dot{\theta}) \sigma_{\Delta t}$. This error provides the uncertainty of the laser shooting in MEMS lidar applications.

2.3 Phase Locked Loop

The mirror can be operated by a phase locked loop (PLL). Typically, PLLs are used in digital circuits to generate an arbitrary frequency from a given reference frequency.

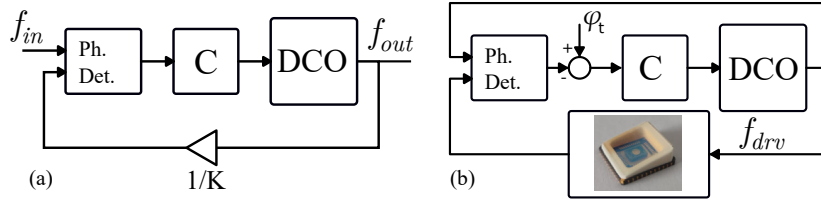


Figure 6: (a) typical PLL used in digital circuits to generate a frequency from a given reference frequency (b) PLL used to drive a mechanical system, such as a MEMS mirror

Figure 6(a) illustrates the PLL used to generate a frequency from a given reference, consisting of a digitally controlled oscillator (DCO), a phase detector, and a controller. The phase detector measures the phase between the reference signal and the output signal and outputs a signal proportional to the phase difference. The phase difference is then used by the controller to calculate the DCO frequency. The divider on the feedback path allows to set the frequency ratio between f_{in} and f_{out} .

The PLL can also be used to drive resonant mechanical systems, such as the MEMS mirror. The mirror is placed on the feedback path, and the phase detector measures the phase difference between the actuation and sensing signal, as shown in Figure 6(b). The target phase can be set with the φ_t input, which allows the definition of the operating point. Assuming that the controller C is properly tuned, the PLL then adjusts the DCO frequency such that the desired target phase is reached. This is important for multiple reasons:

- Operating the mirror at a specific phase implies that the oscillation is always kept.
- If the phase matches the target phase, the laser triggering can be derived from the driving signal.

- As shown in Figure 3, a specific phase corresponds to a specific frequency, and therefore also a specific amplitude. Therefore, the PLL also stabilizes the amplitude.

2.3.1 Linearization at an operation point

Even though the mirror is a nonlinear system, for small changes around a predefined operating point the behavior is linear. This allows the usage of simple controller design methods. The local dynamics are identified directly via measurements, which offers a simpler alternative to the calculation of the local dynamics from the nonlinear dynamics.

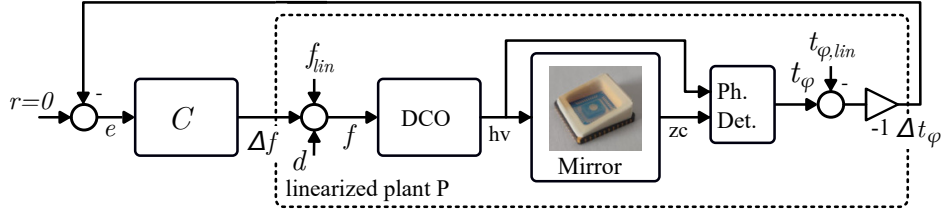


Figure 7: Block diagram of the PLL-operated MEMS mirror.

The DCO, the MEMS mirror, and the phase detector are identified together and result in a plant P , as illustrated in Figure 7. The input of the plant is the frequency Δf and the output is the phase error in time Δt_φ , measured in nanoseconds. The relation between global values and small deviations from them at the operating point is given by

$$f = f_{lin} + \Delta f, \quad (11)$$

$$\Delta t_\varphi = -(t_\varphi - t_{\varphi,lin}), \quad (12)$$

where f is the mirror frequency, t_φ is the phase in time, f_{lin} and $t_{\varphi,lin}$ are the values at the operating point and Δf and Δt_φ are small deviations from the operating point. On the top response curve an increase in frequency results in a decrease of the phase, as shown in Figure 3. When linearizing, this would result in a transfer function where the phase starts at -180° . To obtain a transfer function, where the phase starts at 0° , an inverter is added to the output of the linearized model.

The goal of the PLL is to stabilize the mirror at the operating point, which means that the target phase r is always 0 and the controller input $e = -\Delta t_\varphi$. Disturbances, caused by external shocks or vibrations, can be modeled with a disturbance input d , acting on the input f of the system. The transfer function from d to Δt_φ is called process sensitivity function S_d and is given by

$$S_d = \frac{\Delta t_\varphi}{d} = \frac{P}{1 + CP}. \quad (13)$$

2.3.2 I-Controller Design for the Phase Locked Loop

An I controller is given in the form

$$C_I = \frac{K_I}{s}, \quad (14)$$

or in the discrete domain

$$C_{z,I} = \frac{k_i}{z - 1}, \quad (15)$$

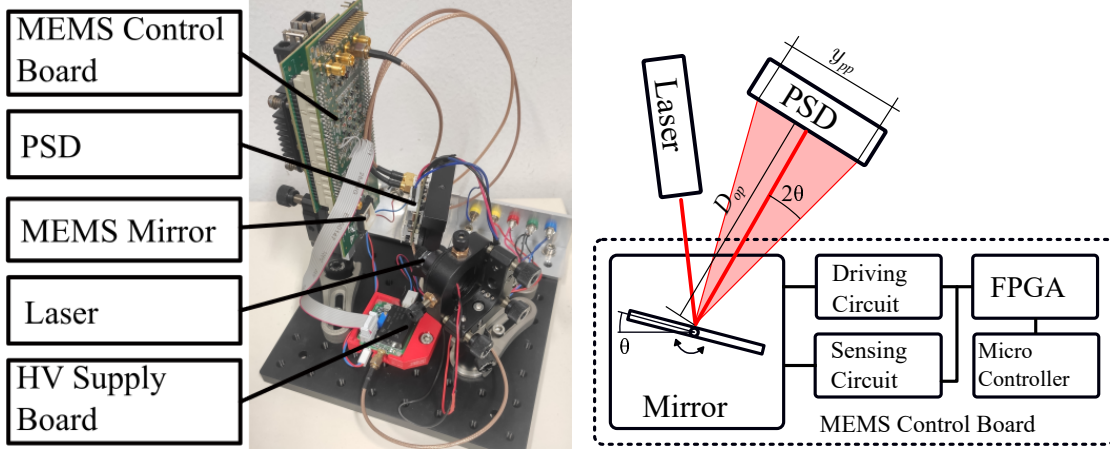


Figure 8: (left) Picture and (right) block diagram of the experimental setup

where K_I and k_i are the i-gains of the controller. The integrating behavior ensures that the error is regulated to 0. For a discrete implementation, the transfer function in z-domain needs to be converted to a state-space representation, i.e.

$$\begin{aligned}\xi_{k+1} &= \xi_k + k_i e_k, \\ \Delta f_k &= \xi_k,\end{aligned}\tag{16}$$

where ξ_k is the state of the controller at time step k .

2.3.3 PI-Controller Design for the Phase Locked Loop

A PI controller is given in the form

$$C_{PI} = \frac{K_I(1 + sK_{PI})}{s},\tag{17}$$

or in the discrete domain

$$C_{z,PI} = k_p + \frac{k_i}{z - 1},\tag{18}$$

where K_I , K_{PI} , k_p and k_i are gains of the controller. Converted to time domain, the discrete PI-controller can be denoted as

$$\begin{aligned}\xi_{k+1} &= \xi_k + k_i e_k, \\ \Delta f_k &= \xi_k + k_p e_k.\end{aligned}\tag{19}$$

3 Experimental Setup

Figure 8 describes a lab setup for the MEMS scanning system, which consists of a MEMS mirror mounted directly to the MEMS control board, a laser and a position-sensitive detector (PSD), which is used to measure the angle of the MEMS mirror.

The following table explains the interface on the side of the setup.

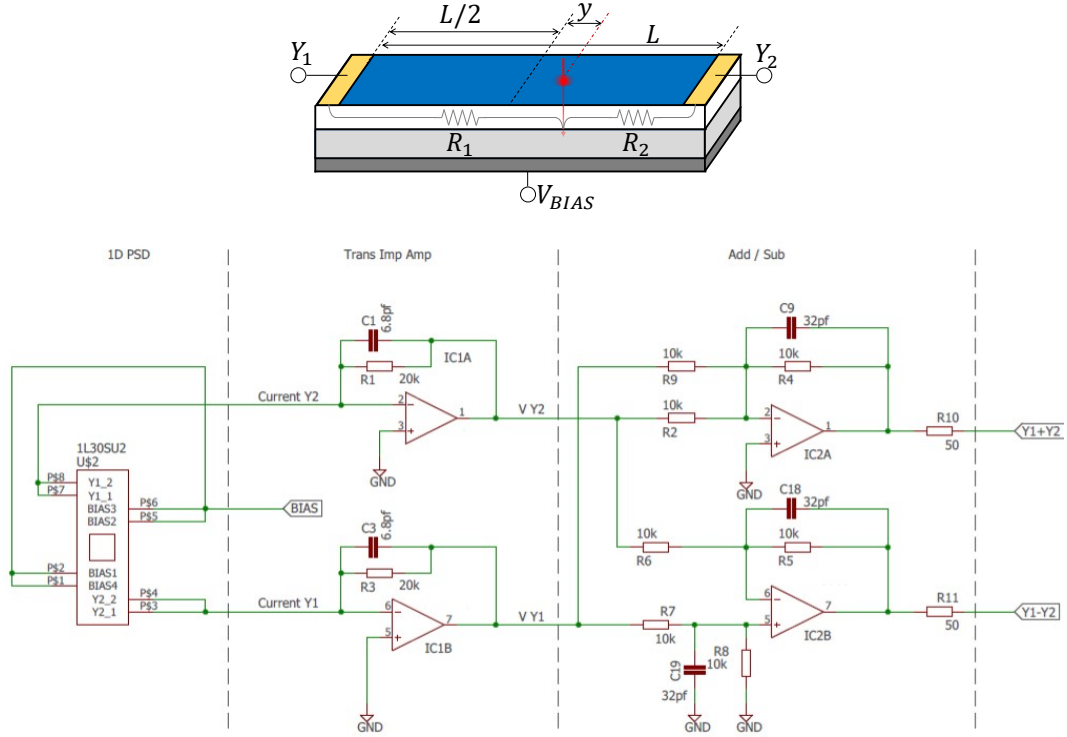


Figure 9: (top) Schematic diagram of a 1D lateral effect PSD with a length of photoactive area L . From the top, it consists of three layers of p-type resistive layer (white), a high resistive intrinsic or low doped layer (bright gray), and n-type layer (dark gray) of common electrode for reverse bias voltage V_{BIAS} . Each end of the p-type resistive layer has electrodes Y_1 and Y_2 that collect divided photocurrent by the distances from the beam position y . (bottom) Schematics of PSD readout circuit.

Label	Type	Direction	default value	Description
V_{+12V}	banana plug	input	12V	supply voltage
GND	banana plug	input	0V	ground
V_{-12V}	banana plug	input	-12V	supply voltage
V_{BIAS}	banana plug	input	2.5V	bias voltage for the PSD
V_{laser}	banana plug	input	2.5V	supply voltage for the laser
D_{act}	BNC connector	output	-	driving signal
V_I	BNC connector	output	-	proportional to comb drive current
V_{diff}	BNC connector	output	-	PSD difference voltage
V_{sum}	BNC connector	output	-	PSD sum voltage

3.1 Position Sensitive Detector Unit

Figure 9 shows a schematic diagram of a PIN junction-based lateral effect PSD with a photoactive area length of $L = 30$ mm. From the top, the first semiconductor device layer (white) is a resistive p-type doped layer, the second layer (bright gray) is an intrinsic or low doped layer with high resistance, and the bottom layer (dark gray) is an n-type layer for the common electrode, which is used for applying reverse bias voltage V_{BIAS} that improves the bandwidth

of PSDs. Each end of the p-type resistive top layer has electrodes Y_1 and Y_2 for collecting photocurrents on both sides of the PSD. When laser light shines a small spot on the active area of the PSD, photocurrent is generated at the local area near the beam position. The photocurrent is divided and flows into two electrodes Y_1 and Y_2 by the resistances R_1 and R_2 , where R_1 and R_2 are proportional to the length of travel to each electrode Y_1 and Y_2 , i.e. $L/2 + y$ and $L/2 - y$, respectively. By accumulating this local current division along the beam intensity distribution, the center of gravity of the beam on the PSD, simply also called as the beam position, is obtained by the measured photo current as

$$y = \frac{L}{2} \frac{I_{Y_2} - I_{Y_1}}{I_{Y_1} + I_{Y_2}} = \frac{L}{2} \frac{V_{\text{diff}}}{V_{\text{sum}}}, \quad (20)$$

where I_{Y_1} and I_{Y_2} denote the current from the electrodes Y_1 and Y_2 . Transimpedance amplifiers change the weak photo currents to voltage signals. Then, an analog adder circuit generates V_{sum} , proportional to $I_{Y_1} + I_{Y_2}$, which represents the intensity information of the light beam. The difference voltage V_{diff} , proportional to $I_{Y_1} - I_{Y_2}$, encodes the beam position. Figure 9 also illustrates an example circuit for this calculation.

Assume that at $\theta = 0$ the reflected beam is orthogonal to the PSD. Then the mirror amplitude a is obtained from the peak-to-peak beam position $y_{pp} = y_{\text{max}} - y_{\text{min}}$ as

$$a = \frac{1}{2} \tan^{-1} \left(\frac{y_{pp}}{2D_{op}} \right), \quad (21)$$

where D_{op} denotes an operation distance from the mirror surface to the PSD and y_{pp} denotes the peak to peak beam position, measured by the PSD. An accurate measurement of the operation distance is important because the uncertainty directly scales the scanning angle calculation. In addition, an external light or diffused laser beam beside the laser beam spot also degrade the accuracy by the increase of V_{sum} by shining the entire sensor surface while there is not much different in V_{diff} , causing shrinks of the value.

3.2 MEMS Control Board

The MEMS control board, engineered by Infineon, operates the mirror. It contains a socket for the mirror itself, analog components to implement the circuits shown in Fig. 5, and a Zynq 7000 device. The Zynq 7000 contains a field programmable gate array (FPGA) and a microcontroller. The FPGA runs with a 100 MHz clock and contains hardware modules of the digitally controlled oscillator (DCO) and the phase detector. The microcontroller configures the hardware modules on the FPGA and implements the rest of the control, e.g. the PLL and user interface. The user interface is implemented by a simple text based menu over a USB connection.

3.3 Operation Tutorial

3.3.1 Open-Loop Operation

The interface is a simple character-based menu navigation, implemented in `menu.c`. Single characters can be sent to the board, which performs the required action, and usually responds by reprinting the menu, showing the available commands. The space character can be used to refresh the menu and does not perform any additional action.

The characters 0, 1 and 2 are used to transition between the modes. 0 enables the **IDLE** mode, i.e. stops the mirror operation. 1 enables the **OPEN-LOOP** mode, and 2 operates the mirror in **CLOSED-LOOP** mode, provided that the PLL is implemented.

In open-loop mode the MEMS mirror is actuated at the frequency displayed in the terminal. The keys **q**, **w**, **e**, **r**, **t** and **z** can be used to change the frequency by different steps, as stated on the serial terminal. The characters **a**, **s**, **d** and **f** change the duty cycle. In closed-loop mode the frequency cannot be changed, since its controlled by the PLL.

The startup procedure is as follows:

1. use 1 to enable the **OPEN-LOOP** mode
2. start with a high frequency of 4.5 kHz.
3. slowly decrease the frequency by pressing **t**
4. when the amplitude jump occurs, increase the frequency again by pressing **w**, reaching even higher amplitudes

3.3.2 Identification of MEMS mirror

An identification by step response is also already provided. The identification is implemented in the function **MirrorId** in **mirror.c** and can be started by sending **i** and **I**. **i** applies a step of -10 Hz on the driving frequency f , and can be used to obtain the plant transfer function. **I** applies an emulated disturbance in form of a step of -10 Hz on d , and can be used to verify the closed-loop behavior of the system. In both cases, the same algorithm is used:

1. Upon receiving of **i** or **I** the control board applies a step on f or d , respectively.
2. The board samples 512 values of the phase t_φ in nanoseconds, and prints them to the serial terminal in form of an array, which can be copied and pasted to MATLAB.
3. When completed, the value of f or d is reset to the previous value.

If the noise of the mirror is large, the step of -10 Hz can be increased in the function **MirrorId**.

3.3.3 Timing Jitter Measurements

The measurement of the noise on t_φ is similar to the identification. By pressing **n** the noise measurement is started, and 256 values of t_φ in nanoseconds are returned. They can then be further evaluated with MATLAB.

4 Exercise Instruction

4.1 Precautions



Failing to respect the following warnings may cause serious damage to the equipment and/or to the setup!

- **High Voltage!**

The MEMS mirror uses high DC voltage up to 100 V. Do not touch any parts of the board when the high-voltage is supplied to the MEMS control board.

- **Electrostatic Discharge!**

The PCB boards such as the MEMS control board, the HV supply board and the PSD board have electronic components, which are very sensitive to electrostatic discharge. Therefore do not touch the PCBs as much as possible and ground yourself electrically by time to time.

- **Do not change the optical alignment!**

If the optical alignment is changed, please report to the tutor to fix it.

- **Class 2 Lasers!**

Do not stare or look directly into the light of the laser.

4.2 Preparation (Homework)

- a. Assume that a state space model of the linearized plant P is given by

$$\begin{aligned}\dot{x} &= \begin{bmatrix} -91 & -307 \\ 256 & 0 \end{bmatrix} x + \begin{bmatrix} 128 \\ 0 \end{bmatrix} \Delta f, \\ \Delta t_\varphi &= [180 \quad 55] x.\end{aligned}\tag{22}$$

Calculate the transfer function in pole, zero and gain representation and draw the Bode plot and the pole-zero plot.

- b. Generate a unit step response with a sampling rate of 4.5 kHz and 512 samples from the system given in (22). Add a Gaussian random noise of $\mathcal{N} \sim (0, 2)$. Use the MATLAB functions `iddata` and `tfest` for system identification and retrieve the continuous time transfer function of the system and compare it to the original model given in (22). Discuss the influence of the amplitude of the step and the amplitude of the noise on the quality of the identified model.
- c. Design I-controllers C_I such that the loop function $L = C_I P$ has a crossover frequency f_c of 100, 200, 300, 400 Hz. What is the phase margin each controller achieves?
- d. Design a PI-controller C_{PI} such that the loop function $L = C_{PI} P$ has a crossover frequency of 400 Hz and a phase margin of exactly 70° . Argue why the I-controller with 400 Hz cannot reach this phase margin. Plot the loop function, controller transfer function, and process sensitivity function S_d for the I-controller with 400 Hz crossover frequency and the PI-controller. Simulate and compare the step responses of the process sensitivity function of the two controllers.
- e. Convert your PI-controller and the plant P to discrete systems with the MATLAB command `c2d`. Simulate a closed-loop step response on the disturbance d of the discrete closed-loop system. Extract the parameters k_i and k_p of (18) from the transfer function in z-domain of your PI-controller.
[Tip] Automate the identification and controller design process, such that you can save time in the lab. Implement MATLAB functions `design_I_cntrl(P, fc)` and `design_PI_cntrl(P, fc, phase_margin)` that return the controllers in continuous domain.
- f. Argue if a PID controller or a PI controller is better to control the system given in (22). Could a PID controller achieve an even higher bandwidth?
- g. Add a discrete implementation for the I-controller and the PI-controller in the supplied C source code by modifying `mirror.c`. The control loop is shown in Figure 7.
- add variables f_{lin} and $t_{\varphi, \text{lin}}$ for the operation point to the `MEMSMirror` struct defined in `Mirror.h`.
 - add variables for the controller state, p-gain, and i-gain to the `MEMSMirror` struct.
 - The function `MirrorInterrupt` is called every time a new DCO period starts. Perform the controller calculations here, use the function `MirrorGetPhase` to get t_φ and `MirrorSetFrequency` to set the new frequency f . Don't forget to consider the operating point.

[Hint] Code section, where your code should be added, are marked with *ToDo*.

4.3 Experiments

In the course of the following experiments, the dynamics of the MEMS mirror are explored, the localized dynamics are identified and a PLL is designed to stabilize the mirror.

Frequency response of a resonant MEMS mirror

1. Operate the mirror in open-loop and use the startup procedure described in Section 3.3.1 to start up the mirror. Observe the difference signal of the PSD V_{diff} and note down the frequency of the amplitude jump and the frequency where the mirror amplitude falls down again.
2. Measure the amplitude of the PSD voltage, the distance from the PSD to the mirror, and calculate the maximum mirror angle.
3. Draw the nonlinear frequency response curve of the mirror, as shown in Figure 3. Set 10-20 frequency points over the serial connection and record the amplitude of V_{diff} , and the phase in time with the oscilloscope. The phase in time is measured as a time difference between the zero-crossing of the mirror and the switching-off of the driving voltage. Draw the frequency-amplitude plot, frequency-phase plot, and phase-amplitude plot.
4. (optional) High order parametric responses. Operate the mirror with 1, and 2/3 of the mechanical oscillation frequency, i.e. 1/2 and 1/3 of the driving frequency of Exercise 1. If an amplitude jump occurred, take a screenshot of the oscilloscope, showing driving signal and mirror amplitude, given by the PSD voltage. A higher duty cycle can help to achieve the amplitude jump.

Current Self-sensing

4. Operate the mirror at high amplitudes, i.e. the phase should be around 5% of the mirror half period. Observe the PSD signal V_{diff} and the comb drive signal V_I to find the region where the zero-crossing of the mirror occurs. Measure the slope of the signal around the zero-crossing with the cursor function of the oscilloscope.
5. Measure the noise of the signal with the measure function (AC-RMS Full Screen) of the oscilloscope. Calculate the expected timing jitter with (6).
6. Measure the timing jitter directly via FPGA by using the procedure described in Section 3.3.3 and calculate the standard deviation. The phase detector on the FPGA uses an advanced algorithm to remove noise, so the timing jitter should be lower the one calculated in the previous exercise.
7. The comb drive signal V_I is only valid if the driving voltage is turned on. To measure the full shape of the comb drive signal, switch off the mirror by typing 0. This keeps the driving voltage high. Capture a single oscilloscope screenshot shortly after switching off, showing the comb drive signal, similar to the one in Figure 4.

Identification of Local Dynamics at Operating Points

8. Select a frequency such that the phase t_φ is around 5% of the mirror half-period. Set the values for f_{lin} and $t_{\varphi,lin}$ at the operating point in `mirror.c` according to your selection. If the mirror is operated on the top response curve, t_φ is displayed at the bottom of the menu.
9. Use the procedure described in Section 3.3.2 to obtain the response of t_φ to a step on f . Calculate Δt_φ from it and identify the continuous 2nd-order model P of the local dynamics with `iddata` and `tfest`. Verify the model by simulating a step response and comparing to the measured data. Plot the transfer function and pole-zero plot of the identified model.

Design of Phase Locked Loop

10. Design a continuous-time I-controller with a crossover frequency of 60 Hz for the PLL, and convert it to a discrete-time controller. Note that the controller runs once per mirror half period, so the sampling frequency equals the chosen frequency at the operating point f_{lin} .
11. Compile and download your implementation from the homework exercise to the micro-controller.
12. Use the time-domain identification provided to measure the response to a emulated disturbance in form of a step on d in closed-loop mode. Compare the measured step response to simulation data to verify your implementation.
13. Design a PI-controller with a crossover frequency of 200 Hz and a phase margin of 70° for the PLL. As before, measure a closed-loop step response on d and compare to simulation. Compare the step-response to the results achieved with pure I-control.
14. Measure the timing jitter in closed loop. Does your PI-controller and I-controller increase the timing jitter compared to the open-loop case? Why?

4.4 Report

Write a report, including the experiment results. The homework must not be included into the report. Also, add the following points:

- Discuss the reason why the backbone curve in Figure 3 is always at the left side of the measured response curve with comb drive actuation. **Hint:** See Eq. (1) and Figure 4.
- Argue why a I-controller with higher gain, or a PID-controller do not achieve better results.
- Compare the MEMS mirror to a driven harmonic oscillator. How would the driving torque τ_{comb} look? How would the response curve change? How does the backbone curve of a linear oscillator look?

References

- [1] H. W. Yoo, N. Druml, D. Brunner, C. Schwarzl, T. Thurner, M. Hennecke, and G. Schitter. MEMS-based lidar for autonomous driving. *Elektrotechnik und Informationstechnik*, 135(6):408–418, 2018.
- [2] David Brunner, Han Woong Yoo, and Georg Schitter. Precise phase control of resonant MOEMS mirrors by comb-drive current feedback. *Mechatronics*, 71:102420, November 2020.